

CCAS 4.3 Donation System

Requirements Specification

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2.0 FUNCTIONAL REQUIREMENTS (MoSCoW METHOD)

2.1 MUST HAVE REQUIREMENTS

ID	Requirement	Description	Acceptance Criteria
REQ_001	Donor Registration and Profile Creation	System must allow donors to register and create profiles.	Donors can create accounts, log in, and update profile information.
REQ_002	Charity Registration and Profile Creation	System must allow charities to register and manage profiles.	Charities can create accounts, provide verification documents, and manage their profile.
REQ_003	Charity and Campaign Browsing/Search	Donors can browse charities and campaigns with search/filter options.	Users can search by charity name, campaign category, location, or keyword.
REQ_004	Donation Processing	Donors can make one-time donations securely.	Donations processed securely with multiple

			payment options.
REQ_005	Donation Tracking and Confirmation	System tracks donations and sends confirmation.	Each donation generates a unique transaction ID and confirmation message.
REQ_006	Campaign Creation and Management	Charities can create and manage campaigns.	Campaigns include title, description, target amount, end date, and progress tracking.
REQ_007	Transparency and Reporting	System provides transparency through reports and updates.	Donors can view impact reports; charities can generate usage reports.

2.2 SHOULD HAVE REQUIREMENTS

ID	Requirement	Description	Acceptance Criteria
REQ_008	Recurring Donations Setup	Donors can set up recurring donations.	Donors can schedule, modify, or cancel recurring donations from dashboard.
REQ_009	Charity Dashboard for Statistics	Charities have a dashboard for donation analytics.	Dashboard displays total donations, campaign performance, donor demographics.
REQ_010	Donor Donation History	Donors can view their	History includes date, amount,

		donation history.	recipient charity, and donation purpose.
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2.3 COULD HAVE REQUIREMENTS

ID	Requirement	Description	Acceptance Criteria
REQ_011	Social Media Sharing	Donors can share donations on social media.	One-click sharing to Facebook, Twitter, Instagram with pre-formatted messages.
REQ_012	Donor Thank-You Notes	Charities can send personalized thank-you notes.	Charity admins can send templated or custom messages via email.

2.4 WON'T HAVE REQUIREMENTS (Current Scope)

ID	Requirement	Description	Reason
REQ_013	Automated Tax Deduction Certificates	System won't handle tax certificates automatically.	Complex legal requirements and jurisdictional variations make automation impractical for initial release.

3.0 NON-FUNCTIONAL REQUIREMENTS

ID	Category	Description	Metrics / Validation
REQ_NF_001	Performance	Handle 100 concurrent users, respond within 2 seconds for 95% of operations.	Load testing with 100+ simulated users.
REQ_NF_002	Security	Encrypt donor payment info,	AES-256 encryption, PCI DSS

		comply with GDPR/CCPA.	compliance, security audit & penetration testing.
REQ_NF_003	Scalability	Scalable to handle 10,000+ users and 50,000+ donation records.	Modular design, cloud-ready infrastructure, database indexing optimization.
REQ_NF_004	Usability	Intuitive UI, mobile-responsive, WCAG 2.1 AA compliance.	User testing, usability surveys.

4.0 USE CASE SCENARIOS

UC_001: Donor Makes One-Time Donation

Primary Actor: Registered Donor

Preconditions:

- Donor is logged into the system
- Donor has selected a specific charity or campaign
- Donor has valid payment method on file

Main Flow:

1. Donor selects donation amount (\$50)
2. System displays donation summary (amount, recipient, processing fee)
3. Donor selects payment method (credit card)
4. Donor enters payment details (card number, expiry, CVV)
5. System validates payment information
6. Donor confirms donation
7. System processes payment through secure gateway
8. System records donation in database
9. System sends confirmation email with receipt
10. System updates campaign total

Postconditions:

- Donation recorded with status 'Completed'
- Donor receives confirmation email with transaction ID

- Campaign total is updated
- Charity notified of new donation

Alternative Flows:

- Payment Declined: System displays error and prompts for alternative payment method
- Insufficient Funds: System notifies donor and suggests lower amount

UC_002: Charity Administrator Creates Campaign

Primary Actor: Charity Administrator

Preconditions:

- Administrator is logged in with charity admin privileges
- Charity profile is verified and active

Main Flow:

11. Administrator navigates to 'Create Campaign'
12. System displays campaign creation form
13. Administrator enters campaign details: Title, Description, Target Amount, End Date, Category
14. System validates form inputs
15. Administrator uploads supporting images/documents
16. System creates campaign with unique ID
17. System publishes campaign to public listings
18. Administrator receives confirmation

Postconditions:

- Campaign is live and visible to donors
- Campaign tracking metrics initialized (current: \$0, progress: 0%)
- Administrator can manage campaign from dashboard

UC_003: Donor Views Donation History

Primary Actor: Registered Donor

Preconditions:

- Donor has made at least one previous donation

Main Flow:

19. Donor logs into account
20. Donor navigates to 'My Donations' section
21. System retrieves donor's donation history from database
22. System displays donations in chronological order
23. Donor filters by date range, charity, or amount
24. Donor selects specific donation for details
25. System displays donation receipt and impact report
26. Donor exports history as PDF (optional)

Postconditions:

- Donor can see complete giving history
- Donor can access tax documentation
- Donor can download receipts for record-keeping

UC_004: Donor Registration and Authentication

Primary Actor: Donor

Secondary Actor: System

Preconditions:

- Donor is not already registered in the system

Main Flow:

27. Donor accesses the registration page
28. Donor enters personal details (name, email, password, contact information)
29. System validates the input data
30. System creates a donor account and stores it securely
31. Donor logs in using registered credentials

Postconditions:

- Donor account is successfully created and authenticated

Alternative Flows:

- If validation fails, the system displays an error and requests correction

UC_005: Browse and Search Charity Campaigns

Primary Actor: Donor

Preconditions:

- System is operational

Main Flow:

32. Donor browses the list of active charity campaigns
33. Donor filters campaigns by category, location, or urgency
34. System displays relevant campaigns with details (goal, description, progress)

Postconditions:

- Donor views available campaigns of interest

UC_006: Make a Donation

Primary Actor: Donor

Secondary Actor: Payment Gateway

Preconditions:

- Donor is logged in
- Campaign is active

Main Flow:

35. Donor selects a charity campaign
36. Donor enters donation amount and payment method
37. System redirects to secure payment gateway
38. Payment gateway processes the transaction
39. System records the donation and updates campaign progress
40. Donor receives confirmation and receipt

Postconditions:

- Donation is successfully recorded and acknowledged

Alternative Flows:

- If payment fails, the system notifies the donor and allows retry

UC_007: Manage Fundraising Campaigns

Primary Actor: Charity Administrator

Preconditions:

- Administrator is authenticated

Main Flow:

41. Administrator creates a new fundraising campaign
42. Administrator defines goal amount, description, timeline, and media
43. System publishes the campaign
44. Administrator monitors donations and campaign progress

Postconditions:

- Campaign is active and visible to donors

UC_008: Track Donation History

Primary Actor: Donor

Preconditions:

- Donor is logged in

Main Flow:

45. Donor accesses donation history page
46. System retrieves and displays past donations with dates, amounts, and campaigns

Postconditions:

- Donor has visibility into personal contribution records

UC_009: Generate Financial and Transparency Reports

Primary Actor: Charity Administrator

Preconditions:

- Donations exist in the system

Main Flow:

47. Administrator selects reporting period and report type
48. System aggregates donation and campaign data
49. System generates downloadable reports (e.g., totals, allocation, progress)

Postconditions:

- Transparency and accountability reports are generated

UC_010: Manage Charity Events

Primary Actor: Charity Administrator

Preconditions:

- Administrator is authenticated

Main Flow:

50. Administrator creates a charity event
51. Administrator sets date, location, description, and target audience
52. System publishes the event
53. Donors can view and register for the event

Postconditions:

- Event is available for donor participation

UC_011: Receive Notifications and Updates

Primary Actor: Donor

Preconditions:

- Donor is subscribed to notifications

Main Flow:

54. System detects campaign updates or event announcements
55. System sends notifications via email or in-app alerts

Postconditions:

- Donor stays informed about supported causes

UC_012: Admin User Management

Primary Actor: System Administrator

Preconditions:

- System administrator is authenticated

Main Flow:

- 56. Administrator manages user roles (donor, charity admin)
- 57. Administrator activates, suspends, or removes accounts
- 58. System enforces access control rules

Postconditions:

- User access and roles are correctly managed

UC_012: Audit and Compliance Monitoring

Primary Actor: System Administrator

Preconditions:

- Donation and transaction data exists

Main Flow:

- 59. Administrator reviews system logs and transaction records
- 60. System highlights inconsistencies or anomalies
- 61. Administrator exports audit reports if required

Postconditions:

- System compliance and data integrity are ensured